1 $\sqrt{(-12)^2 + 5^2}$ or $\sqrt{144 + 25}$ **[M1]**

13 **[A1]**

[Total Marks: 2]

2 $\sqrt{5^2 + (-12)^2}$ or $\sqrt{25 + 144}$ **[M1]**

13 **[A1]**

[Total Marks: 2]

3 (a) Determines column vector $\mathbf{a} = \begin{pmatrix} -6 \\ 8 \end{pmatrix}$ **[M1]**

10 **[A1]**

(b) Determines column vector $\mathbf{b} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ **[B1]**

Calculates resultant vector $2\begin{pmatrix} -6 \\ 8 \end{pmatrix} + 3\begin{pmatrix} 3 \\ -4 \end{pmatrix} = \begin{pmatrix} -12 \\ 16 \end{pmatrix} + \begin{pmatrix} 9 \\ -12 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ **[M1]**

5 **[A1]**

[Total Marks: 5]

4 (a) $\sqrt{(-8)^2 + 15^2}$ or $\sqrt{64 + 225}$ **[M1]**

17 **[A1]**

(b) $\mathbf{p} + 2\mathbf{q} = \begin{pmatrix} -8 \\ 15 \end{pmatrix} + \begin{pmatrix} 4 \\ -8 \end{pmatrix} = \begin{pmatrix} -4 \\ 7 \end{pmatrix}$ **[M1]**

$\sqrt{(-4)^2 + 7^2}$ or $\sqrt{16 + 49}$ **[M1]**

$\sqrt{65}$ **[A1]**

[Total Marks: 5]

5 (a) $\begin{pmatrix} 6 \\ 8 \end{pmatrix}$ **[B1]**

(b) $|\vec{AB}| = \sqrt{6^2 + 8^2} = \sqrt{36 + 64}$ **[M1]**

10 **[A1]**

(c) (i) $|\mathbf{v}| = \sqrt{(-12)^2 + (-5)^2} = \sqrt{144 + 25}$ **[M1]**

13 **[A1]**

(ii) $\vec{AB} + \mathbf{v} = \begin{pmatrix} 6 \\ 8 \end{pmatrix} + \begin{pmatrix} -12 \\ -5 \end{pmatrix} = \begin{pmatrix} -6 \\ 3 \end{pmatrix}$ **[M1]**

$\sqrt{(-6)^2 + 3^2} = \sqrt{45} = 3\sqrt{5}$ **[A1]**

[Total Marks: 7]

6 **(a)** Sets up equation: $p^2 + p^2 = 18$ **[M1]**

3 **[A1]**

(b) Finds expression for $\mathbf{w}$: $\begin{pmatrix} 4-t \\ 1+2t \end{pmatrix}$ **[M1]**

Forms magnitude squared equation: $(4-t)^2 + (1+2t)^2 = 29$ **[M1]**

Expands brackets correctly: $16 - 8t + t^2 + 1 + 4t + 4t^2 = 29$ **[M1]**

Rearranges to a 3-term quadratic: $5t^2 - 4t - 12 = 0$ **[M1]**

$t = 2$, or $t = -1.2$ (or $-\frac{6}{5}$) **[A1]**

[Total Marks: 7]

7 **(a)** $\sqrt{x^2 + (x+2)^2} = 10$ or $x^2 + (x+2)^2 = 10^2$ **[M1]**

$x^2 + x^2 + 4x + 4 = 100$ **[M1]**

$2x^2 + 4x - 96 = 0$ leading to $x^2 + 2x - 48 = 0$ with no errors seen **[A1]**

(b) $(x+8)(x-6) = 0$ or correct substitution into quadratic formula **[M1]**

$x = -8, x = 6$ **[A1]**

(c) $\mathbf{u} = \begin{pmatrix} -8 \\ -6 \end{pmatrix}$ **[B1]**

$\mathbf{u} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$ **[B1]**

[Total Marks: 7]

8 **(a)** **(i)** $\sqrt{6^2 + 2^2} = \sqrt{40} = 2\sqrt{10}$ **[B1]**

(ii) $\vec{AB} = \begin{pmatrix} -2 \\ 6 \end{pmatrix} - \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} -8 \\ 4 \end{pmatrix}$ **[M1]**

$|\vec{AB}| = \sqrt{(-8)^2 + 4^2} = \sqrt{80} = 4\sqrt{5}$ **[A1]**

(b) $|\vec{OB}| = \sqrt{(-2)^2 + 6^2} = \sqrt{40} = 2\sqrt{10}$ **[M1]**

$|\vec{OA}| = |\vec{OB}| = 2\sqrt{10}$, so triangle OAB is isosceles **[A1]**

$|\vec{OA}|^2 + |\vec{OB}|^2 = 40 + 40 = 80 = |\vec{AB}|^2$, so angle $AOB = 90°$ **[A1]**

(c) **(i)** $\vec{OP} = \vec{OA} + \frac{3}{4}\vec{AB} = \begin{pmatrix} 6 \\ 2 \end{pmatrix} + \frac{3}{4}\begin{pmatrix} -8 \\ 4 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 0 \\ 5 \end{pmatrix}$ **[A1]**

(ii) $\vec{PQ} = \begin{pmatrix} 8 \\ m \end{pmatrix} - \begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 8 \\ m-5 \end{pmatrix}$ **[M1]**

$8^2 + (m-5)^2 = 17^2$ giving $(m-5)^2 = 225$ **[M1]**

$m - 5 = -15$ (since $m < 0$), so $m = -10$ **[A1]**

[Total Marks: 11]

9 **(a)** $\begin{pmatrix}-5\\12\end{pmatrix}$ **[B1]**

(b) $\sqrt{(-5)^2+12^2}$ or $\sqrt{25+144}$ **[M1]**

13 **[A1]**

(c) $3^2+c^2=13^2$ or $9+c^2=169$ **[M1]**

$c^2=160$ **[M1]**

12.6 and -12.6 (or $\pm 4\sqrt{10}$) **[A1]**

[Total Marks: 6]

10 **(a)** $\vec{AC}=\begin{pmatrix}7-3\\1-(-2)\end{pmatrix}$ or $\begin{pmatrix}4\\3\end{pmatrix}$ **[M1]**

5 **[A1]**

(b) $\vec{AB}=\begin{pmatrix}-8\\y+2\end{pmatrix}$ **[B1]**

$\sqrt{(-8)^2+(y+2)^2}=17$ **[M1]**

$64+(y+2)^2=289$ or $(y+2)^2=225$ **[M1]**

$y+2=15$ or $y+2=-15$ **[M1]**

$y=13$, $y=-17$ **[A1]**

[Total Marks: 7]

11 **(a)** Expresses vector $\vec{PQ}$ as $\begin{pmatrix}k-2\\5-(-3)\end{pmatrix}=\begin{pmatrix}k-2\\8\end{pmatrix}$ **[M1]**

Sets up magnitude equation: $(k-2)^2+8^2=10^2$ **[M1]**

Simplifies to solve for k: $(k-2)^2=36$ **[M1]**

$k=8$, or $k=-4$ **[A1]**

(b) Finds $\vec{QR}$ using $Q(8,5)$ and $R(4,1)$: $\vec{QR}=\begin{pmatrix}4-8\\1-5\end{pmatrix}=\begin{pmatrix}-4\\-4\end{pmatrix}$ **[M1]**

Calculates magnitude: $\sqrt{(-4)^2+(-4)^2}=\sqrt{32}$ **[M1]**

5.66 **[A1]**

[Total Marks: 7]

12 **(a)** $\begin{pmatrix}9-2\\4-(-3)\end{pmatrix}$ **[M1]**

$\begin{pmatrix}7\\7\end{pmatrix}$ **[A1]**

(b) $\sqrt{7^2+7^2}$ or $\sqrt{98}$ **[M1]**

9.90 **[A1]**

(c) $\sqrt{k^2+4^2}=5$ or $k^2+4^2=5^2$ **[M1]**

$k^2=25-16$ or $k^2=9$ **[M1]**

$k=3$, $k=-3$ **[A1]**

[Total Marks: 7]

13 (a) $2\mathbf{p}-\mathbf{q}=\begin{pmatrix}8\\-6\end{pmatrix}-\begin{pmatrix}-1\\5\end{pmatrix}=\begin{pmatrix}9\\-11\end{pmatrix}$ **[M1]**

$|2\mathbf{p}-\mathbf{q}|=\sqrt{9^2+(-11)^2}=\sqrt{202}$ **[M1]**

14.2 **[A1]**

(b) $|\mathbf{p}|=\sqrt{4^2+(-3)^2}=5$ **[B1]**

$k=-\dfrac{35}{5}=-7$ **[M1]**

$\begin{pmatrix}-28\\21\end{pmatrix}$ **[A1]**

(c) $(t+3)^2+(2t-1)^2=34$ **[M1]**

$t^2+6t+9+4t^2-4t+1=34$ simplifying to $5t^2+2t-24=0$ **[M1]**

$(5t+12)(t-2)=0$ **[M1]**

$t=2$ and $t=-2.4$ (or $-\dfrac{12}{5}$) **[A1]**

[Total Marks: 10]

14 (a) $(k+3)^2+(2k-1)^2=98$ **[M1]**

$k^2+6k+9+4k^2-4k+1=98$ **[M1]**

$5k^2+2k+10=98$ leading to $5k^2+2k-88=0$ with no errors seen **[A1]**

(b) $(5k+22)(k-4)=0$ or $\dfrac{-2\pm\sqrt{2^2-4(5)(-88)}}{2(5)}$ **[M2]**

If M0, allow [M1] for a correct partial factorisation e.g. $(5k+a)(k+b)$ where $ab=-88$ or correct substitution into quadratic formula with one sign error

4 and -4.4 **[A1]**

(c) $\begin{pmatrix}7\\7\end{pmatrix}$ **[B1]**

[Total Marks: 7]

15 (a) $\mathbf{u}+\mathbf{v}=\begin{pmatrix}3k+1\\k-1\end{pmatrix}$ **[M1]**

$(3k+1)^2+(k-1)^2=(\sqrt{50})^2$ **[M1]**

$9k^2+6k+1+k^2-2k+1=50$ **[M1]**

$10k^2+4k+2=50 \implies 5k^2+2k-24=0$ **[A1]**

(b) $(5k + a)(k + b) = 0$ where $ab = -24$ or $5b + a = 2$, or correct substitution into quadratic formula **[M1]**

$(5k + 12)(k - 2) = 0$ **[M1]**

$k = 2$ **[A1]**

$k = -2.4$ or $-\dfrac{12}{5}$ **[A1]**

(c) Substituting $k = 2$ and $k = -2.4$ into $\begin{pmatrix} 2k \\ k - 3 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 4 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -4.8 \\ -5.4 \end{pmatrix}$ **[A1]**

[Total Marks: 10]